SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED
CO1: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)

Assessment Tool Weightage: 30%
Dr Dumpala Prasanth

QUESTIONS: 10Total Marks: 50
Annexure A
Case StudyDuration: 2hrs

Question 1

A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. (5 Marks)

Answer:

<u>MDP Formulation</u>

The warehouse can be modeled as an MDP where the Environment is the warehouse floor plan containing shelves, obstacles, charging stations, and packing stations. The State includes the robot's current coordinates and orientation, battery level, the item(s) currently held, positions of nearby obstacles or other robots (from LiDAR/camera sensors), and the target shelf or packing location. The Action space is discrete/continuous and includes move (forward, backward, turn), pick item, place item, and wait/yield. The Policy maps each state to the action that maximizes expected long-term reward, e.g. a neural network policy trained via Deep Q-Learning or PPO that outputs the next movement or manipulation action.

<u>Reward Function Design</u>

A balanced reward function combines three weighted terms: R = w1*R_speed + w2*R_accuracy + w3*R_safety. R_speed is a small negative step-penalty per time unit encouraging shorter paths, plus a bonus for completing a task before a time threshold. R_accuracy gives a large positive reward for correctly picking/placing the requested item and a large negative penalty for wrong-item selection or damaged goods. R_safety imposes a heavy penalty for collisions with shelves, humans, or other robots, and a smaller penalty for entering restricted/near-miss zones, with weights tuned so that w3 dominates to guarantee safety is never sacrificed for speed.

<u>Analysis and Evaluation</u>

Reward design directly shapes behavior: if the speed term is over-weighted, the robot may learn to cut corners, causing collisions or dropped items; if accuracy is over-weighted without a time penalty, the robot may move unnecessarily slowly. A well-balanced, safety-dominant reward avoids these local optima and produces a robot that is fast, precise, and safe. Reward shaping (e.g., potential-based shaping toward the goal shelf) can accelerate learning without changing the optimal policy, and periodic evaluation using metrics such as pick-success rate, average cycle time, and collision count should guide iterative reward tuning.

Question 2

A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. (5 Marks)

Answer:

MDP Formulation

The Environment is the streaming platform and its catalog. The State representation is a feature vector combining the user's recent viewing/rating history, genre preferences, time of day/device context, and an embedding of previously recommended items (to avoid repetition). The Action space is the selection of the next movie/show (or slate of items) to recommend from the catalog, often reduced via candidate generation to a manageable set. The Policy is typically a contextual-bandit or deep RL policy (e.g., DQN or Actor-Critic) mapping the current user-state to a ranked list of recommendations.

Reward Function Design

The reward combines immediate engagement signals and longer-term satisfaction: a small positive reward for a click, a larger reward for watching a significant fraction of the content (e.g., >70% completion), a strong positive reward for an explicit like/favorite, and negative rewards for skips, early exits, or 'not interested' feedback. To avoid short-term clickbait optimization, a delayed component such as next-session return rate or weekly watch-time can be added.

Analysis and Evaluation

Reward signal choice strongly affects engagement quality: optimizing purely for clicks encourages sensational/clickbait-like recommendations, while rewarding completion and return visits promotes genuinely satisfying recommendations. The exploration-exploitation trade-off is critical—pure exploitation of known preferences causes a filter bubble and stale recommendations, while excessive exploration frustrates users with irrelevant content. An epsilon-greedy or Thompson-sampling strategy with a decaying exploration rate, combined with diversity-aware reward bonuses, balances discovery of new content against personalization accuracy.

Question 3

An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. (5 Marks)

Answer:

MDP Formulation

The Environment is the farm field with its crop, soil, and local climate system. The State includes soil moisture and nutrient levels (from IoT sensors), weather forecast data (rainfall probability, temperature, humidity), crop growth stage, and historical irrigation/fertilizer application. The Action space includes discrete or continuous choices such as irrigate (amount/duration), apply fertilizer (type/quantity), or take no action. The Reward is computed from crop health indicators, water/fertilizer usage efficiency, and eventually the harvested yield and quality.

Reward Function Design

A practical reward combines an immediate shaping term (e.g., keeping soil moisture and nutrients within an optimal band, penalizing over- or under-watering/fertilizing to discourage resource waste) with a large terminal reward tied to final crop yield and quality at harvest. Penalties are added for excessive water/fertilizer consumption to encourage sustainability.

Analysis and Evaluation

Because crop yield is only observable months after an irrigation/fertilizer decision, this is a classic delayed-reward problem: the agent must learn to associate early-season actions with a distant outcome, which increases variance and slows convergence. Algorithms that handle long horizons well—such as Deep Q-Networks with experience replay, or Actor-Critic methods with eligibility traces/n-step returns—are well suited here because they propagate delayed reward signals more efficiently than simple Monte Carlo methods, and using immediate proxy rewards (moisture/nutrient bands) as shaping terms helps the agent learn useful behavior long before harvest.

Question 4

A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. (5 Marks)

Answer:

MDP / Environment Modeling

The Environment is the live network traffic stream. The State is constructed from features extracted per time-window or per-connection: packet rate, protocol type, source/destination entropy, failed login attempts, port-scan indicators, and statistical deviations from a learned baseline of normal traffic. The State must be updated continuously as new traffic flows in, making this a streaming, partially observable environment since the true intent of traffic (malicious or benign) is never fully known.

Action and Reward Design

Actions range from passive monitoring/logging, raising an alert, throttling suspicious traffic, to blocking/isolating a source IP. Rewards give a strong positive value for correctly detecting and blocking an actual attack, a strong negative value for a false positive (blocking legitimate traffic, which harms business operations), a smaller negative value for a missed attack (false negative), and a small negative step cost for unnecessary alerts to discourage over-triggering.

Challenges and Evaluation

Two major challenges dominate this setting: sparse rewards, since actual attacks are rare compared to benign traffic, meaning the agent receives very little positive/negative signal to learn from, and real-time decision-making, since the agent must act within milliseconds without waiting to fully verify an attack. These are mitigated using reward shaping with intermediate anomaly-scores, prioritized experience replay to over-sample rare attack events, and hybrid systems where a fast-reacting policy triggers containment while a slower model performs deeper verification, evaluated via precision/recall and response latency rather than raw accuracy.

Question 5

A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. (5 Marks)

Answer:

MDP Formulation

The Environment is the city's road network with real-time supply (available drivers) and demand (ride requests). The State captures driver locations, current demand density per zone, time of day, weather, and traffic conditions. The Action space includes dispatching a specific driver to a rider, repositioning idle drivers toward high-demand zones, and setting a surge-pricing multiplier for a zone/time window.

Reward Function Design

The reward balances multiple stakeholders: positive reward for completed trips (weighted by fare and reduced rider wait-time), a bonus for efficient driver utilization (minimizing idle time), and a penalty for excessive surge pricing that could be seen as exploitative or that reduces trip completion rate due to riders abandoning the app.

Analysis and Evaluation

Environmental uncertainty—unpredictable traffic, weather-driven demand spikes, and cancellations—makes the transition dynamics highly stochastic, requiring the policy to generalize robustly rather than overfit to average conditions; techniques like robust RL or ensemble-based value estimation help. Ethically, dynamic pricing raises fairness concerns: aggressive surge pricing during emergencies or in low-income areas can be exploitative or discriminatory, so the reward and policy should include fairness constraints (e.g., price caps, transparency, and equitable service-level guarantees across neighborhoods) evaluated alongside profit metrics.

Question 6

A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. (5 Marks)

Answer:

MDP Formulation

The Environment is the home's physical and sensor infrastructure. The State includes occupancy detection per room, current temperature/humidity/light levels, outdoor weather, time of day, and user-set preferences or manual overrides. The Action space includes adjusting HVAC setpoints, turning lights on/off or dimming, and switching modes (eco/comfort/away).

Reward Function Design

The reward is a weighted combination of comfort and energy terms: R = w1*R_comfort - w2*R_energy_cost, where R_comfort rewards keeping temperature/light within the user's preferred comfort band (with penalties for occupied-but-uncomfortable states), and R_energy_cost penalizes electricity consumption, especially during peak-price periods.

Analysis and Evaluation

Comfort and energy-saving are inherently conflicting objectives—maximum comfort typically demands more energy—so the weights w1 and w2 must be tunable (or learned via multi-objective/Pareto-based RL) to reflect user priorities, and constraints (e.g., never letting temperature drop below a safety threshold) should be enforced regardless of reward weighting. Reward shaping, such as giving small intermediate rewards for anticipatory pre-cooling before occupancy, improves sample efficiency, but poorly designed shaping can bias the agent toward the shaping signal rather than the true objective, so shaping terms must be validated to not alter the optimal policy (potential-based shaping is preferred).

Question 7

A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance. (5 Marks)

Answer:

MDP Formulation

The game is modeled as an MDP (or a Markov Game in the multi-agent/adversarial setting) where the Environment is the full game engine and opponent. The State is the complete or partially observed game board/unit configuration, resources, and turn history. The Action space includes all legal moves available at a given game state, which can be very large (combinatorial) in strategy games.

Reward Structure for Long-Term Strategy

To encourage long-term strategy rather than short-sighted tactics, the reward should be primarily sparse and outcome-based (large positive reward for winning, large negative for losing, zero/small for a draw), supplemented with carefully designed shaping rewards for strategic sub-goals (territory control, resource accumulation, unit advantage) that correlate with long-term winning probability without dominating the final-outcome signal.

Analysis and Evaluation

Because the state-action space is huge, exploration strategy critically affects performance: pure random exploration rarely discovers winning strategies, so techniques like Monte Carlo Tree Search combined with policy/value networks (as in AlphaZero-style self-play), or curiosity-driven exploration bonuses for novel states, are used to explore promising strategic lines efficiently. Self-play with a growing pool of past policy versions also prevents overfitting to a single opponent style, and performance should be evaluated using win-rate against a diverse set of benchmark opponents rather than a single fixed adversary.

Question 8

A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. (5 Marks)

Answer:

MDP Formulation

The Environment is the stream of financial transactions. The State is a feature vector per transaction (or per account over a time window): transaction amount, merchant category, location, device fingerprint, time since last transaction, and deviation from the account's typical spending pattern. The Action space includes approve, flag for manual review, or block/decline the transaction.

Reward Function Design

The reward gives a strong positive value for correctly blocking/flagging genuine fraud, a strong negative value for a false positive (blocking a legitimate customer transaction, which damages trust and revenue), and a large negative value for a false negative (an undetected fraudulent transaction resulting in financial loss), typically scaled by the transaction amount to reflect real monetary risk.

Challenges and Evaluation

Fraudulent transactions are a very small fraction of all transactions (severe class imbalance), so the agent receives very few positive fraud-detection learning signals; this is addressed with techniques like prioritized experience replay, reward rescaling for rare events, or synthetic oversampling of fraud-like patterns during training. Delayed feedback is also a challenge since some frauds are confirmed only days later via chargebacks, requiring the system to update its policy asynchronously as ground-truth labels arrive. Because incorrect decisions carry real financial and reputational risk, evaluation must go beyond accuracy to precision/recall on the minority class, financial loss avoided, and customer friction (false-positive rate), with conservative deployment (human-in-the-loop review) for high-value transactions.

Question 9

A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. (5 Marks)

Answer:

MDP Formulation

The Environment is the patient's ongoing rehabilitation process. The State includes the patient's current physical condition/mobility scores, pain levels, exercise history and adherence, and progress metrics from wearable sensors or clinician assessments. The Action space is the selection of the next exercise, its intensity, duration, and rest periods. The Policy adapts the exercise plan session-by-session based on the patient's evolving state.

Reward Function Design

The reward combines short-term signals (successful completion of a session without excessive pain or injury risk, adherence to the plan) with longer-term signals (measurable improvement in mobility/strength scores over successive weeks), with a strong negative penalty for any action that risks re-injury or exceeds a clinically safe intensity threshold.

Analysis, Ethics, and Evaluation

Rewards tied to recovery outcomes are inherently delayed—true improvement is only observable over weeks—so the agent must handle long-horizon credit assignment, typically via discounted returns with intermediate proxy rewards (session completion, self-reported pain reduction) to guide learning before full recovery is observed. Patient variability (age, condition severity, comorbidities) means a single global policy will not generalize well, so the system should use patient-context conditioning or hierarchical/personalized RL rather than a one-size-fits-all policy. Ethically, safety must always dominate the reward function (hard constraints, not just penalties, on unsafe intensities), informed consent and clinician oversight are required, and the system should be evaluated on safety incidents and clinician-validated outcomes, not purely on the RL agent's self-reported reward.

Question 10

A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. (5 Marks)

Answer:

MDP Formulation

The Environment is the electricity distribution grid comprising generation sources, storage, and consumer demand. The State includes current load per grid segment, generation output (including variable renewable sources), storage/battery levels, weather forecasts, and time-of-day demand patterns. The Action space includes adjusting power distribution/routing, charging or discharging storage, triggering demand-response signals to consumers, and activating/deactivating peaking generation units.

Reward Function Design

The reward balances efficiency and reliability: R = w1*R_efficiency - w2*R_cost - w3*R_reliability_penalty, where R_efficiency rewards minimizing transmission losses and matching supply closely to demand, R_cost penalizes the use of expensive peaking power or excessive storage cycling, and R_reliability_penalty is a heavy penalty for brownouts, blackouts, or failure to meet demand at any node.

Analysis and Evaluation

Scalability is a central challenge because a real grid has thousands of interconnected nodes, making the joint state-action space intractable for a single centralized agent; this is addressed with multi-agent RL, where local agents control sub-regions and coordinate through shared signals or a hierarchical controller, decomposing the problem while preserving global stability. Real-time learning is also difficult because grid dynamics change continuously (weather, demand surges) and mistakes can cause real outages, so safe/constrained RL approaches that guarantee hard reliability constraints are preferred over unconstrained exploration, with performance evaluated on peak-load reduction, cost savings, and reliability metrics (e.g., SAIDI/SAIFI) rather than reward alone.

Code of Conduct:

I G.Dhanusha Sree(192325012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G.Dhanusha Sree

RUBRICS FOR CASESTUDY

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand